DSSSB • TGT COMPUTER SCIENCE • POST 41/26

Part B
Computer Science & teaching

Core CS,mathematics,business,physics and teaching methodology, mapped to the supplied subject headings.

30 days | 3–4 hours daily | English explanations

357 worked questions • Prepared 20 October 2026

How to use: Attempt every question in a type before reading its solutions. Cover the answers on paper, or use the answer toggle in the HTML workbook. Redo errors after 1, 3 and 7 days.

Coverage: One section is provided for each mapped syllabus/question family. Each section places practice questions first, then worked answers and a shortcut or solving rule. Each regular Part A/B section now includes two new questions marked Additional practice. Original pattern-based expected practice is included throughout; it is informed practice, not a prediction or a claim to reproduce every historical question.

Syllabus status: The 2026 advertisement verifies the exam scheme for 41/26. The supplied 56/24 subject syllabus is used provisionally: a separate 41/26 subject-syllabus PDF was not verified. Follow any later official 41/26 notice. Exact exam date was not supplied; timetable days are relative.

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B01 • ORIGINAL PRACTICE

Computer fundamentals, units and devices

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q105-106,124-126; P34 · TGT Computer Science 2021 file 2 (2021) Q105; P37 · TGT Computer Science 2021 file 5 (2021) Q122-124

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B01.1 How many bits are in 4 KiB?

B01.2 Which device reads shaded bubbles on an answer sheet?

B01.3 Match nano, pico and femto to powers of seconds.

B01.4 Additional practice: How many bits are in 2 KiB, using binary units?

B01.5 Additional practice: Which memory is volatile: SRAM, ROM or an SSD?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

A bit is binary; 1 byte=8 bits. KiB=1024 bytes; KB in SI=1000 bytes. OCR reads characters, OMR detects marks, MICR reads magnetic-ink characters. Diligence is repeated work without fatigue.

B01.1 32,768. 4×1024×8. Keep bits and bytes distinct.

B01.2 OMR. OCR would recognise printed character shapes.

B01.3 10^-9,10^-12,10^-15 seconds. Each is 1000 times smaller than the previous one.

B01.4 16,384 bits. 2×1024 bytes×8 bits/byte. Distinguish KiB (1024 bytes) from decimal kB (1000 bytes).

B01.5 SRAM. It requires power to retain stored bits. “Static” means no periodic refresh while powered, not nonvolatile storage.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B02 • ORIGINAL PRACTICE

Number bases and complements

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q108,115; P34 · TGT Computer Science 2021 file 2 (2021) Q107; P37 · TGT Computer Science 2021 file 5 (2021) Q120,131

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B02.1 Convert decimal 45 to binary and hexadecimal.

B02.2 Find the 8-bit two’s-complement representation of -5.

B02.3 Convert $(132)_4$ to decimal.

B02.4 Additional practice: Convert $(101101)_2$ to decimal.

B02.5 Additional practice: Represent −18 in 8-bit two's complement.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Expand in positional powers. For binary↔octal group 3 bits; binary↔hex group 4. n-bit two’s-complement range is -2^(n-1) to 2^(n-1)-1. Invert bits and add 1 to negate.

B02.1 101101_2 and $2D_{16}$. 45=32+8+4+1; group binary as 0010 1101.

B02.2 11111011. +5=00000101; invert→11111010; add 1.

B02.3 30. 1×16+3×4+2. Digits in base b must be below b.

B02.4 45. Add weights 32+8+4+1. Write powers of two only under the 1 bits.

B02.5 11101110. +18=00010010; invert to 11101101 and add 1. Retain exactly 8 bits.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

Codes, parity and representation

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q121-122; P37 · TGT Computer Science 2021 file 5 (2021) Q129

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B03.1 Write decimal 59 in 8421 BCD.

B03.2 Find the even-parity bit for data 1011001.

B03.3 Convert binary 1011 to Gray code.

B03.4 Additional practice: An even-parity word has data bits 1011011. What parity bit is appended?

B03.5 Additional practice: Convert binary 10110 to reflected Gray code.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

ASCII is a character code; BCD encodes each decimal digit separately; Gray code changes one bit between adjacent codewords. An even-parity bit makes the total count of ones even.

B03.1 0101 1001. Encode 5 then 9; ordinary binary for 59 is different.

B03.2 0. There are already four ones.

B03.3 1110. Keep first bit 1; XOR each neighbouring binary pair: $1 \oplus 0 = 1, 0 \oplus 1 = 1, 1 \oplus 1 = 0$.

B03.4 1. The data contains five 1s, an odd count; add a 1 to make the total six. Parity detects odd numbers of flipped bits but cannot locate a bad bit.

B03.5 11101. Keep the first bit; each subsequent Gray bit is XOR of adjacent binary bits: $1, (1 \oplus 0), (0 \oplus 1), (1 \oplus 1), (1 \oplus 0)$.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B04 • ORIGINAL PRACTICE

Boolean algebra and universal gates

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q128-134; P34 · TGT Computer Science 2021 file 2 (2021) Q127,131

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B04.1 Simplify A+AB+A′B.

B04.2 Express AND using only 2-input NOR gates.

B04.3 An alarm activates if any of three sensors is 1. Which gate implements this?

B04.4 Additional practice: Simplify A+AB.

B04.5 Additional practice: Express NOT A using only a two-input NAND gate.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Absorption: A+AB=A; A(A+B)=A. De Morgan: (AB)′=A′+B′ and (A+B)′=A′B′. NAND and NOR are universal. A⊕A=0; A⊕0=A.

B04.1 A+B. A+AB=A; A+A′B=(A+A′)(A+B)=A+B.

B04.2 A AND B = NOR(NOR(A,A),NOR(B,B)). Three gates: invert both inputs then NOR.

B04.3 OR. AND would require all three; XOR would reject some multi-sensor cases.

B04.4 A, by absorption. Factor A(1+B)=A. A truth table is a reliable fallback if the identity is unfamiliar.

B04.5 NAND(A,A). Its output is ¬(A·A)=¬A. Tie both inputs together; the same construction with NOR also inverts.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B05 • ORIGINAL PRACTICE

K-maps, combinational circuits and multiplexers

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q131,133; P34 · TGT Computer Science 2021 file 2 (2021) Q128,130,138

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B05.1 Simplify F(A,B,C)=Σm(0,1,2,3).

B05.2 How many select lines does a 16-to-1 multiplexer need?

B05.3 For a half subtractor performing A-B, give difference and borrow.

B05.4 Additional practice: Simplify F(A,B,C)=Σm(0,1,2,3), with A the most significant bit.

B05.5 Additional practice: A 16-to-1 multiplexer needs how many select inputs?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Group K-map ones in powers of 2; wrapping is allowed, diagonal adjacency is not. A 2^n-input multiplexer needs n select lines. Half adder sum=A⊕B, carry=AB.

B05.1 A′. All four combinations with A=0 are included; B and C vary away.

B05.2 4. 2^4=16.

B05.3 Difference=A⊕B; borrow=A′B. Borrow occurs only at A=0,B=1.

B05.4 A'. All four minterms have A=0 while B and C vary. A group of four eliminates two variables.

B05.5 4. A select word of s bits chooses among 2^s inputs, so s=$\log_2 16$. Data-input count and select-input count are different.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B06 • ORIGINAL PRACTICE

Flip-flops, counters and sequential circuits

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q132; P34 · TGT Computer Science 2021 file 2 (2021) Q129,139; P37 · TGT Computer Science 2021 file 5 (2021) Q127

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B06.1 What happens to a JK flip-flop when J=K=1 at the active clock edge?

B06.2 How many flip-flops are minimally required for a modulo-10 counter?

B06.3 A 5-stage Johnson counter has how many states in its normal cycle?

B06.4 Additional practice: What does a JK flip-flop do on its active clock edge when J=K=1?

B06.5 Additional practice: How many flip-flops are minimally required for a modulo-10 binary counter?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

A flip-flop stores one bit. Combinational outputs depend on present inputs; sequential circuits also depend on state. Some sequential circuits are asynchronous, so “all need a clock” is too broad.

B06.1 It toggles: next state is the complement of current state.

B06.2 4. Need ceil($\log_2 10$)=4, giving up to 16 states.

B06.3 10. Johnson=2n; ordinary n-stage one-hot ring=n; binary counter=2^n.

B06.4 Toggles its prior state. Do not confuse this with the invalid input of a basic SR latch.

B06.5 4. Find the smallest n with 2^n≥10; 3 bits provide only 8 states. Additional logic limits the counter to ten states.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B07 • ORIGINAL PRACTICE

Architecture: memory, address lines and cache

Observed: P37 · TGT Computer Science 2021 file 5 (2021) Q139; cache is syllabus extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B07.1 A memory has 32K words of 16 bits. How many address and data lines?

B07.2 A direct-mapped cache has 64 lines,16-byte blocks and 16-bit byte addresses. Find tag,index,offset bits.

B07.3 Cache hit time is 2 ns, hit ratio 0.9, and additional miss penalty 50 ns. Find average time.

B07.4 Additional practice: A byte-addressable memory has 20 address lines. What is the maximum directly addressable capacity?

B07.5 Additional practice: A cache hits 90% of the time. Hit time is 2 ns and a miss adds 50 ns. Find average access time.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

For W words of b bits, address lines=$\log_2 W$ when W is a power of 2; data lines=b. Cache effective time must match the stated convention: a miss penalty can include or exclude cache lookup.

> **B07.1** 15 address lines and 16 data lines. 32K=2^15 words.

> **B07.2** Offset=4,index=6,tag=6. Total=16.

> **B07.3** 7 ns. Hit time+(miss rate×extra penalty)=2+0.1×50.

> **B07.4** 2^20 bytes=1 MiB. Address lines count locations; multiply by bytes per location only if the memory is not byte-addressable.

> **B07.5** 7 ns. AMAT=hit time+miss rate×miss penalty=2+0.1×50. The wording says the penalty is additional time.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B08 • ORIGINAL PRACTICE

Architecture: addressing, CPU and performance

Observed: P37 · TGT Computer Science 2021 file 5 (2021) Q140-143; performance is syllabus extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B08.1 A CPU executes 10^6 instructions at average CPI 2 and 1 GHz. Find CPU time.

B08.2 How many memory accesses obtain an operand through single-level memory-indirect addressing, excluding instruction fetch?

B08.3 A program spends 40% of time in a part sped up 2×. Overall speedup?

B08.4 Additional practice: A program executes 10^6 instructions with average CPI 2 on a 1 GHz processor. Find CPU time.

B08.5 Additional practice: Which addressing mode uses the operand value stored directly inside the instruction?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

PC holds the next instruction address; IR holds the current instruction. Register addressing takes the operand from a register; indirect addressing obtains an address from another location. CPU time=instruction count×CPI/clock rate.

B08.1 0.002 s=2 ms. 2×10^6/10^9.

B08.2 Two: read effective address, then read operand.

B08.3 1/(0.6+0.4/2)=1.25. Amdahl’s law applies to the original time fractions.

B08.4 0.002 seconds=2 ms. Time=instructions×CPI/clock frequency. Convert GHz to 10^9 cycles/s first.

B08.5 Immediate addressing. Distinguish it from direct addressing, where the instruction supplies the memory address of the operand.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B09 • ORIGINAL PRACTICE

C: operators, division and output tracing

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q115; P34 · TGT Computer Science 2021 file 2 (2021) Q115; P37 · TGT Computer Science 2021 file 5 (2021) Q114

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B09.1 For int a=9,b=4, find a/b and a%b.

B09.2 Evaluate -11%4 in C99 and later.

B09.3 Does printf("%d %d",i++,i++) have one portable, defined result in C?

B09.4 Additional practice: In C99 or later, what are −17/5 and −17%5 for integer operands?

B09.5 Additional practice: Is printf("%d", i++ + ++i) a portable way to obtain one numeric result in C?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Trace one statement at a time with a variable table. Integer division truncates toward zero in modern C. Remainder has the dividend's sign. Do not assign a numerical answer to undefined behaviour.

> **B09.1** 2 and 1. Both operands are integers.

> **B09.2** -3. Quotient=-2, and -11=(-2)×4+(-3).

> **B09.3** No. Modifications of i are unsequenced relative to one another; behaviour is undefined. Do not rely on one compiler run.

> **B09.4** −3 and −2. Division truncates toward zero; remainder satisfies a=(a/b)b+a%b. Avoid importing Python's floor-division behaviour into C.

> **B09.5** No. The unsequenced modifications of i cause undefined behaviour. Do not force an arithmetic answer for an undefined expression.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B10 • ORIGINAL PRACTICE

C: arrays, pointers, strings and allocation

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q110; P37 · TGT Computer Science 2021 file 5 (2021) Q110,117; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q123

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B10.1 int a[]={4,8,12}; int *p=a; What is *(p+2)?

B10.2 How many char elements are needed for the C string "CODE"?

B10.3 Ignoring padding, a struct contains char[3] and an int of 4 bytes. Is its actual size guaranteed to be 7 bytes?

B10.4 Additional practice: For int a[]={10,20,30,40}; int *p=a+1;, what is *(p+2)?

B10.5 Additional practice: In C, what is sizeof("cat") and why?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Array indices start at zero in C. Pointer arithmetic scales by pointed-to type. A C string needs a terminating null byte. malloc allocates uninitialised storage; calloc zero-initialises allocated bytes.

B10.1 12. p+2 points to the third int, not two bytes later.

B10.2 Five, including the terminating null character.

B10.3 No. Actual layout can include padding for alignment. Seven is only the sum of member sizes.

B10.4 40. p points to a[1], so p+2 points to a[3]. Pointer arithmetic advances by element size, not by a single raw byte.

B10.5 4 bytes. The string literal contains c,a,t and a terminating null character. strlen("cat") is 3.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B11 • ORIGINAL PRACTICE

C: control flow, recursion and preprocessing

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q118; P34 · TGT Computer Science 2021 file 2 (2021) Q112,116

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B11.1 How many body executions for for(int i=1;i<=16;i*=2)?

B11.2 For f(0)=1 and f(n)=n*f(n-1), find f(4).

B11.3 For #define SQR(x) x*x, what does SQR(2+3) expand to and evaluate as?

B11.4 Additional practice: A function returns 1 for n=0 and n*f(n−1) otherwise. What is f(5)?

B11.5 Additional practice: How many times does for(int i=1;i<16;i*=2) execute its body?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Count loop bodies separately from condition checks. A recursive function needs a base case and progress toward it. Macros are textual substitution; parenthesise arguments and the whole expansion.

B11.1 Five: i=1,2,4,8,16. The condition is checked six times.

B11.2 24. Unwind 4×3×2×1×1.

B11.3 2+3*2+3=11. Safer mathematical macro: ((x)*(x)); repeated side effects still need caution.

B11.4 120. It computes factorial: 5×4×3×2×1. Identify the recurrence and base case before expanding every call.

B11.5 4 times, for i=1,2,4,8. Doubling loops require counting powers rather than assuming n iterations.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B12 • ORIGINAL PRACTICE

C++ and object-oriented concepts

Observed: P37 · TGT Computer Science 2021 file 5 (2021) Q109; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q110,150

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B12.1 Same method name with different parameter lists: overloading or overriding?

B12.2 A base pointer calls an overridden virtual method in a derived object. Which implementation runs?

B12.3 Which C++ mechanism frees resources when an automatic object leaves scope?

B12.4 Additional practice: A base pointer refers to a derived C++ object. Which declaration enables a derived override to be called through that pointer?

B12.5 Additional practice: Why should a polymorphic C++ base class usually have a virtual destructor if objects will be deleted through a base pointer?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Encapsulation bundles state and behaviour; abstraction exposes essentials; inheritance extends a type; polymorphism allows a common interface with different behaviour. Overloading differs from overriding.

> **B12.1** Overloading. It is normally resolved at compile time.

> **B12.2** Derived implementation, through runtime dispatch.

> **B12.3** Its destructor; RAII ties resource lifetime to object lifetime.

> **B12.4** Declare the base member function virtual. Dynamic dispatch uses the actual object type for virtual calls; ordinary nonvirtual calls follow static type rules.

> **B12.5** To ensure destruction dispatches correctly to the derived object. Without the required virtual destructor, deleting a derived object through such a base pointer has undefined behaviour.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B13 • ORIGINAL PRACTICE

Java: compilation, types and exceptions

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q155-157; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q132,152

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B13.1 What is produced when a normal Java source class is compiled?

B13.2 Two separately created String objects contain "abc". Which normally tests equal contents: == or equals?

B13.3 Which handles an exception: try, catch or finally?

B13.4 Additional practice: Does Java's == compare the content of two distinct String objects?

B13.5 Additional practice: If a Java method throws an unhandled checked exception, what must its declaration normally include?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Java source compiles to bytecode executed by a JVM. == on object references tests identity; equals may test logical equality. Java is always pass-by-value, including copies of reference values.

> **B13.1** JVM bytecode in a .class file.

> **B13.2** equals. == tests whether both references designate the same object.

> **B13.3** catch handles a matching exception. finally normally performs cleanup; it is not guaranteed after forced JVM termination.

> **B13.4** No; it compares references. Use equals for String content equality, with appropriate null handling.

> **B13.5** A compatible throws declaration. Alternatively the method must catch the checked exception. Unchecked RuntimeException subclasses follow different compile-time rules.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B14 • ORIGINAL PRACTICE

Python essentials and problem solving

Observed: P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q160,188

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B14.1 What is [10,20,30,40][1:3]?

B14.2 After d={"x":2}; d["x"]=7, how many entries are present?

B14.3 What prints: a=[1,2]; b=a; b.append(3); print(a)?

B14.4 Additional practice: What are Python's −7//3 and −7%3?

B14.5 Additional practice: For a=[1,2]; b=a; b.append(3), what is a?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Python lists are mutable, tuples immutable. Dictionary keys are unique. Slices exclude the stop index. Distinguish identity (is) from equality (==). Python appears in the reviewed 2023 paper.

B14.1 [20,30]. Start included, stop excluded.

B14.2 One. Assignment replaces the value for the existing key.

B14.3 [1,2,3]. a and b reference the same mutable list.

B14.4 −3 and 2. Floor division rounds toward negative infinity; −7=(−3)×3+2. The modulo result has the divisor's sign when nonzero.

B14.5 [1,2,3]. Both names reference the same list. Use a.copy() or slicing for a shallow list copy if aliasing is unwanted.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B15 • ORIGINAL PRACTICE

Stacks, queues, linked lists and expressions

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q117; P37 · TGT Computer Science 2021 file 5 (2021) Q111,113

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B15.1 Evaluate postfix 5 2 3 * +.

B15.2 Convert (A+B)*C to postfix.

B15.3 Enqueue A,B,C; dequeue twice; enqueue D. What is at the front?

B15.4 Additional practice: Convert A+B*C to postfix.

B15.5 Additional practice: Which data structure supports breadth-first traversal: a stack or a queue?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Stack=LIFO; queue=FIFO. Postfix evaluation: pop right operand first, then left. A linked-list insertion is O(1) only if the needed node/location is already available.

B15.1 11. First 2×3=6, then 5+6.

B15.2 AB+C*. Postorder of the expression tree gives postfix.

B15.3 C. FIFO removes A then B.

B15.4 ABC*+. Multiplication has higher precedence, so emit B C * before the final addition with A.

B15.5 A queue. Enqueue discovered vertices and remove from the front; a stack naturally supports depth-first exploration.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B16 • ORIGINAL PRACTICE

Trees, BSTs and traversals

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q117,174,177; P34 · TGT Computer Science 2021 file 2 (2021) Q118,175

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B16.1 Insert 8,3,10,1,6 into an empty BST. Give inorder traversal.

B16.2 For root A with left child B and right child C, give preorder and postorder.

B16.3 In the first BST, delete 8 using its inorder successor. What becomes the root?

B16.4 Additional practice: Insert 40,20,60,10,30 into a BST. What is its inorder traversal?

B16.5 Additional practice: A full binary tree has 9 internal nodes. How many leaves does it have?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Preorder=root-left-right; inorder=left-root-right; postorder=left-right-root. Inorder of a distinct-key BST is sorted. A node with two children may be replaced by its inorder successor or predecessor.

B16.1 1,3,6,8,10.

B16.2 Preorder A,B,C; postorder B,C,A.

B16.3 10. It is the smallest key in 8’s right subtree.

B16.4 10,20,30,40,60. Inorder of a BST lists keys in nondecreasing order when the duplicate convention is respected.

B16.5 10. In a full binary tree every internal node has exactly two children; leaves=internal nodes+1. This does not hold for every arbitrary binary tree.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B17 • ORIGINAL PRACTICE

Heaps, AVL trees and hashing

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q171-174; P37 · TGT Computer Science 2021 file 5 (2021) Q174

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B17.1 Insert 30,20,10 into an AVL tree. Which repair is required?

B17.2 For a max-heap, must a left child exceed the right child?

B17.3 Hash h(k)=k mod 7. Insert 10,17,24 using linear probing into an empty table indexed 0-6. Slots?

B17.4 Additional practice: Insert 30,20,10 into an AVL tree. Which rotation restores balance?

B17.5 Additional practice: With h(k)=k mod 7 and linear probing, place 10,17,24 into an initially empty table.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

A heap is a complete tree with a parent-child ordering property, not a fully sorted tree. AVL balance factors are -1,0,+1. Hashing offers expected O(1) under suitable assumptions, not guaranteed worst-case O(1).

B17.1 A right rotation at 30 (LL imbalance). Root becomes 20.

B17.2 No. Only each parent must be at least as large as its children.

B17.3 3,4,5. All initially hash to 3; probe to the next free slot.

B17.4 A right rotation at 30, an LL case. The resulting root is 20 with children 10 and 30.

B17.5 Slots 3,4,5 respectively. All hash to 3; linear probing advances to the next free slot, wrapping if necessary.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B18 • ORIGINAL PRACTICE

Searching, sorting and complexity

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q111-112,116; P34 · TGT Computer Science 2021 file 2 (2021) Q114; P37 · TGT Computer Science 2021 file 5 (2021) Q112

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B18.1 Maximum comparisons for conventional successful/unsuccessful binary search among 1,000 entries?

B18.2 How many adjacent swaps sort [3,1,2] increasingly?

B18.3 Find complexity: for i=1..n, for j=1..i, constant work.

B18.4 Additional practice: What is the worst-case complexity of binary search in a sorted random-access array?

B18.5 Additional practice: Which standard sort is stable: selection sort or merge sort with ties taken from the left run first?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Compare growth rates: log n < n < n log n < n^2 < 2^n. Binary search needs sorted data. Standard bubble/insertion/selection worst cases are quadratic; mergesort is n log n; quicksort can be quadratic.

B18.1 At most floor($\log_2 1000$)+1=10 key comparisons under the usual one-comparison-per-iteration model.

B18.2 Two, equal to the inversion count: (3,1),(3,2).

B18.3 $\Theta(n^2)$. Work=1+2+...+n=n(n+1)/2.

B18.4 O(log n). Each comparison discards about half the remaining interval. Sorted order and efficient access to the middle are required.

B18.5 The described merge sort. Ordinary selection sort is not stable; implementation details such as tie handling matter.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B19 • ORIGINAL PRACTICE

Graphs and algorithm choices

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q175; P37 · TGT Computer Science 2021 file 5 (2021) Q175,177-178; recurrence added for coverage

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B19.1 An undirected graph has degree sum 18. How many edges?

B19.2 Which finds shortest paths by number of edges in an unweighted graph: BFS or DFS?

B19.3 Edges A-B=2,B-C=3,A-C=8. Shortest A-to-C distance?

B19.4 For T(n)=2T(n/2)+n, give asymptotic time.

B19.5 Additional practice: Which shortest-path method works for nonnegative edge weights: Dijkstra's algorithm or an arbitrary DFS?

B19.6 Additional practice: A connected undirected graph has 8 vertices. How many edges does any spanning tree contain?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Undirected degree sum=2E. BFS finds fewest-edge paths in unweighted graphs. Dijkstra requires nonnegative edge weights; Bellman-Ford handles negative edges and detects reachable negative cycles. MST differs from shortest path.

B19.1 Nine. Handshaking lemma.

B19.2 BFS. It expands distance layers.

B19.3 Five via B.

B19.4 $\Theta(n \log n)$. Each recursion level costs n and there are $\log_2 n$ levels.

B19.5 Dijkstra's algorithm. For graphs with negative weights choose a method whose assumptions allow them, such as Bellman-Ford, and check for reachable negative cycles.

B19.6 7. A tree on V vertices has $V-1$ edges. The original graph may contain more edges.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B20 • ORIGINAL PRACTICE

DBMS: models, ER diagrams and keys

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q139; P37 · TGT Computer Science 2021 file 5 (2021) Q133,135-137

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B20.1 A table has 80 rows and 6 columns. Give cardinality and degree.

B20.2 Each student belongs to one department, and a department has many students. Relationship?

B20.3 If student_id alone uniquely identifies rows, is {student_id,name} a candidate key?

B20.4 Additional practice: What is a candidate key?

B20.5 Additional practice: In an ER model, how is a multivalued attribute commonly represented in Chen notation?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Cardinality of a relation=rows; degree=columns. A candidate key is a minimal superkey. Primary key is the selected candidate key; a foreign key references a candidate/unique key as constrained by the DBMS.

B20.1 Cardinality=80; degree=6.

B20.2 Department-to-student is one-to-many.

B20.3 No: it is a superkey but not minimal.

B20.4 A minimal set of attributes that uniquely identifies each row. A superkey may contain extra attributes; “minimal” means no attribute can be removed while preserving uniqueness.

B20.5 A double oval. A double rectangle denotes a weak entity, so first identify whether the symbol refers to an attribute or an entity.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B21 • ORIGINAL PRACTICE

Functional dependencies and normalisation

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q136; dependency calculations are extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B21.1 R(A,B,C), key AB, dependency A→C. Which normal form is violated if R is in 1NF?

B21.2 R(A,B,C), key A, A→B and B→C, with B not a key and C non-prime. Violation?

B21.3 Given A→B and B→C, find A+ over attributes A,B,C.

B21.4 Additional practice: R(A,B,C) has FDs A→B and B→C. If A is the only candidate key, which dependency causes a 3NF problem?

B21.5 Additional practice: Under A→B and B→C, find A+.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

1NF: atomic values in the relational design sense. 2NF removes partial dependency of non-prime attributes on candidate keys. 3NF handles transitive dependency; formally each nontrivial X→A has X superkey or A prime. BCNF requires every determinant to be a superkey.

B21.1 2NF. Non-prime C depends on part of the key.

B21.2 3NF because of the transitive dependency A→B→C.

B21.3 {A,B,C}. Add B, then C until no more attributes can be added.

B21.4 B→C. B is not a superkey and C is nonprime. This transitive dependency suggests decomposing into (A,B) and (B,C), with the usual lossless/dependency checks.

B21.5 {A,B,C}. Start with A, repeatedly add the right side of any FD whose left side is already included, and stop when no new attribute is added.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B22 • ORIGINAL PRACTICE

SQL: selection, aggregation, NULL and joins

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q140; P37 · TGT Computer Science 2021 file 5 (2021) Q138; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q166

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B22.1 Table T(x) has 2,NULL,5. Find COUNT(*),COUNT(x),SUM(x).

B22.2 Write SQL to list departments with over five employees in Emp(dept,id).

B22.3 What does SELECT * FROM T WHERE x = NULL return under standard SQL NULL semantics?

B22.4 A left table has three rows, only one matching the right table exactly once. How many rows after a LEFT JOIN?

B22.5 Additional practice: A column contains 10,NULL,20. What do COUNT(*),COUNT(column) and AVG(column) return?

B22.6 Additional practice: Which clause filters groups after aggregation: WHERE or HAVING?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

WHERE filters rows before grouping; HAVING filters groups. COUNT(*) counts rows; COUNT(column) ignores NULL. Use IS NULL, not =NULL. An inner join requires matches; a left join preserves left rows.

B22.1 3,2,7.

B22.2 SELECT dept FROM Emp GROUP BY dept HAVING COUNT(*) > 5;

B22.3 No rows satisfy it as true. Use WHERE x IS NULL to select NULL values.

B22.4 Three. Unmatched right-side fields become NULL.

B22.5 3,2,15. COUNT(*) counts rows; ordinary COUNT(column) and AVG ignore NULL inputs.

B22.6 HAVING. WHERE filters individual rows before grouping. This logical ordering explains why a group-total condition belongs in HAVING.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B23 • ORIGINAL PRACTICE

Transactions, concurrency and indexing

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q137-138; P34 · TGT Computer Science 2021 file 2 (2021) Q132-135; ACID is extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B23.1 A transfer debits one account but crashes before crediting another. Which ACID property requires all-or-nothing?

B23.2 Which anomaly occurs when a transaction reads another’s uncommitted update?

B23.3 For range queries on ordered keys, which is generally more suitable: a B+ tree or a hash index?

B23.4 Additional practice: Two transactions hold different resources and each waits for the other's locked resource. What condition exists?

B23.5 Additional practice: In ACID, what does durability mean?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

ACID: atomicity, consistency, isolation, durability. A dirty read sees uncommitted data. B+ tree leaves hold indexed entries and support ranges. A sparse index does not store one entry for every search-key occurrence.

> **B23.1** Atomicity.

> **B23.2** Dirty read.

> **B23.3** B+ tree. Hashing is typically strongest for equality lookup.

> **B23.4** Deadlock. The wait-for graph has a cycle. Prevention, detection and recovery are distinct ways of handling it.

> **B23.5** A committed transaction's effects survive subsequent failures as guaranteed by the system's persistence model. Atomicity concerns all-or-nothing completion; do not interchange them.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B24 • ORIGINAL PRACTICE

Operating systems: processes, threads and scheduling

Observed: P37 · TGT Computer Science 2021 file 5 (2021) Q157-161; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q112

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B24.1 FCFS jobs all arrive at t=0, in order A(4),B(3),C(2). Find average waiting time.

B24.2 For the same jobs under nonpreemptive SJF, find average waiting time.

B24.3 What technique reduces starvation from low priority?

B24.4 Additional practice: Nonpreemptive FCFS schedules jobs arriving at time 0 with bursts 4,3,2 in that order. Find average waiting time.

B24.5 Additional practice: What is the key scheduling difference between a process and its threads?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Process states commonly include ready, running, blocked/waiting. Threads share a process address space but have their own registers, stack and execution state. Waiting time=turnaround-burst; turnaround=completion-arrival.

B24.1 (0+4+7)/3=11/3 time units.

B24.2 Order C,B,A; waits 0,2,5; average=7/3.

B24.3 Aging: gradually increase priority of long-waiting jobs.

B24.4 (0+4+7)/3=11/3 time units. Waiting time excludes the job's own execution.

B24.5 Threads in one process share its address space/resources but have their own execution state such as registers and stacks. This enables communication but requires synchronisation.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B25 • ORIGINAL PRACTICE

Operating systems: paging and replacement

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q158,164; P34 · TGT Computer Science 2021 file 2 (2021) Q154

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B25.1 32-bit virtual address,4 KiB pages. How many page-number and offset bits?

B25.2 With 3 initially empty frames, reference string 1,2,3,1,4,2: FIFO page faults?

B25.3 Using LRU for the same string, how many faults?

B25.4 Which algorithm can show Belady's anomaly: FIFO or LRU?

B25.5 Additional practice: With 3 initially empty frames, how many FIFO faults occur for 1,2,3,1,4,1?

B25.6 Additional practice: For 16-bit virtual addresses and 1 KiB pages, how many page-number bits are there?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Page offset bits=$\log_2$(page size in bytes). FIFO removes the oldest loaded page; LRU removes the least recently used; optimal removes the page whose next use is farthest away. A hit does not advance FIFO loading order.

B25.1 Offset=12; virtual page number=20.

B25.2 Four. Load 1,2,3; hit 1; replace oldest 1 with 4; hit 2.

B25.3 Five. Before 4 the least recent is 2, so replace 2; the final 2 faults.

B25.4 FIFO. More frames can sometimes increase faults; LRU has the stack property.

B25.5 5. First 1,2,3 fault; the next 1 hits; 4 replaces oldest 1; the final 1 faults. A hit does not refresh FIFO insertion order.

B25.6 6. Page offset=$\log_2 1024$=10 bits; page number=16−10=6. Use the virtual-address width asked in the question.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B26 • ORIGINAL PRACTICE

Operating systems: fragmentation and deadlocks

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q159-163; deadlock is extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B26.1 A process needs 18 KiB and gets five 4 KiB pages. Internal waste?

B26.2 Two nonadjacent free holes of 10 and 12 KiB cannot serve a contiguous 18 KiB request. What problem?

B26.3 Resources must always be requested in one global order. Which deadlock condition does this prevent?

B26.4 Additional practice: A 4 KiB fixed allocation unit holds 3,000 bytes. How many bytes of internal fragmentation remain?

B26.5 Additional practice: Name the four necessary Coffman conditions for deadlock.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Internal fragmentation is waste inside allocations; external is scattered free space between allocations. Deadlock needs mutual exclusion, hold-and-wait, no preemption and circular wait. An unsafe state is not necessarily already deadlocked.

B26.1 2 KiB. Allocated 20 minus used 18.

B26.2 External fragmentation.

B26.3 Circular wait.

B26.4 1,096 bytes. Allocated 4096−used 3000. Internal waste lies within an allocated block.

B26.5 Mutual exclusion, hold-and-wait, no preemption and circular wait. Breaking any one prevents a deadlock of this resource-allocation model.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B27 • ORIGINAL PRACTICE

Operating systems: disks, files and system calls

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q152,155-157; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q133,181

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B27.1 Head at 50; FCFS requests 10,70,30. Total head movement?

B27.2 For the same requests under SSTF, choose nearest at each step; if tied choose smaller track. Total movement?

B27.3 Which is a system-call category: file management or font decoration?

B27.4 Additional practice: Disk requests 20,70,40 are served by FCFS from head position 50. Find total movement.

B27.5 Additional practice: Which system call family concept creates a new process in a traditional Unix model: fork or read?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Seek time moves the head to a track; rotational latency waits for the sector. Spooling queues work for a device. System calls request kernel services. For disk algorithms state initial direction and end-track policy.

B27.1 40+60+40=140 tracks.

B27.2 100. 50→30 (20),→10 (20),→70 (60).

B27.3 File management. Opening, reading and closing files require OS services.

B27.4 |50−20|+|20−70|+|70−40|=110 cylinders. Follow the service order and sum absolute differences.

B27.5 fork. read obtains data from a file descriptor; exec replaces a process image rather than creating an additional process by itself.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B28 • ORIGINAL PRACTICE

Networks: OSI, devices and topology

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q120-123,168-171; P37 · TGT Computer Science 2021 file 5 (2021) Q165-170

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B28.1 At which OSI layer is routing normally placed?

B28.2 Which layer provides end-to-end transport and port addressing?

B28.3 A star network's central switch fails. What is the usual effect?

B28.4 Additional practice: Which OSI layer performs routing between networks?

B28.5 Additional practice: In a simple full mesh of 6 nodes, how many point-to-point links are required?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

OSI low to high: Physical,Data link,Network,Transport,Session,Presentation,Application. Switches typically forward frames using MAC addresses; routers route packets using IP. A device can implement multiple layers.

B28.1 Network layer, layer 3.

B28.2 Transport layer, layer 4.

B28.3 Connected nodes lose communication through that switch. A single peripheral cable failure usually affects only that node.

B28.4 Network layer, layer 3. A conventional router forwards IP packets at this layer; a layer-2 switch primarily forwards frames using MAC addresses.

B28.5 15. Every unordered pair needs one link: $n(n-1)/2$. Count links once, not once from each endpoint.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B29 • ORIGINAL PRACTICE

TCP/IP, ports and reliability

Observed: P37 · TGT Computer Science 2021 file 5 (2021) Q167; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q169,182

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B29.1 Match HTTPS,SSH,DNS to 443,22,53.

B29.2 Which TCP messages establish a normal connection?

B29.3 Which protocol supplies error/control messages such as those used by ping?

B29.4 Can UDP never be used for reliable applications?

B29.5 Additional practice: Match default service ports: DNS, HTTPS, SSH.

B29.6 Additional practice: What does TCP's three-way handshake normally exchange?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

TCP is connection-oriented and provides reliable ordered byte streams; UDP supplies datagrams without those guarantees. HTTP=80,HTTPS=443,DNS=53,SSH=22,SMTP=25,FTP control=21. DNS uses both UDP and TCP.

B29.1 HTTPS=443; SSH=22; DNS=53.

B29.2 SYN → SYN-ACK → ACK.

B29.3 ICMP. It is not the application-level mail protocol.

B29.4 False. Reliability can be implemented above UDP; UDP itself does not provide it.

B29.5 DNS 53, HTTPS 443, SSH 22. DNS commonly uses both UDP and TCP; the service and transport must be distinguished.

B29.6 SYN, SYN-ACK, ACK. It synchronises connection state and initial sequence numbers; do not count it as application data delivery.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B30 • ORIGINAL PRACTICE

IP addressing and subnetting

Syllabus extension: computer networks and TCP/protocols

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B30.1 For 192.168.10.70/26, find network and broadcast.

B30.2 How many conventional usable hosts in /27?

B30.3 Split a /24 network into four equal subnets. New prefix?

B30.4 Additional practice: For 192.168.10.64/26, give the network, broadcast and conventional usable host range.

B30.5 Additional practice: How many /27 subnets fit in a /24 network?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

IPv4 has 32 bits. For an ordinary /p subnet with network and broadcast reserved, usable hosts=2^(32-p)-2; /31 point-to-point and /32 are exceptions. Subnet block size in a changing octet=256-mask octet.

B30.1 Network 192.168.10.64; broadcast 192.168.10.127. /26 gives blocks of 64.

B30.2 30. Five host bits give 32 addresses minus 2.

B30.3 /26. Borrow $\log_2 4 = 2$ bits.

B30.4 Network .64, broadcast .127, usable .65–.126. /26 gives a block size of 64 and 6 host bits, hence 62 conventional hosts.

B30.5 8. Three bits are borrowed: 2^(27−24)=8. Each /27 has 32 addresses, ordinarily 30 usable host addresses.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B31 • ORIGINAL PRACTICE

Transmission, switching and error detection

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q161,163-165; P37 · TGT Computer Science 2021 file 5 (2021) Q121,169-170

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B31.1 Send 1,000 bytes over 1 Mbps. Transmission delay, ignoring overhead?

B31.2 A walkie-talkie allows one speaker at a time in either direction. Mode?

B31.3 An undetected error changes two bits under a single parity check. Is this possible?

B31.4 Additional practice: A noiseless 3 kHz channel uses 4 signal levels. What is the Nyquist maximum bit rate?

B31.5 Additional practice: What is the chief purpose of a CRC in ordinary data-link use?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Transmission delay=L/R; propagation delay=distance/signal speed. Simplex is one-way; half-duplex alternates directions; full-duplex supports both simultaneously. CRC detects errors; it does not encrypt.

B31.1 8 ms. 8000 bits/1,000,000 bits per second.

B31.2 Half-duplex.

B31.3 Yes. Even numbers of bit flips preserve parity.

B31.4 12,000 bit/s. $2B \log_2 M = 2 \times 3000 \times 2$. Shannon capacity uses signal-to-noise ratio and answers a different model.

B31.5 Detect transmission errors. It is not encryption, authentication or a general method for correcting every detected error.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B32 • ORIGINAL PRACTICE

Network security and cryptography

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q170; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q130,146

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B32.1 Which security goal detects unauthorised modification?

B32.2 Alice signs a message. Which key does Bob use to verify it?

B32.3 What is the main purpose of a salt in password hashing?

B32.4 An email mimics a bank and requests credentials. Attack type?

B32.5 Additional practice: A digital signature is generated with which key, and verified with which key?

B32.6 Additional practice: A fake login page steals a password. Which attack is this?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

CIA=confidentiality,integrity,availability. Hashing is not encryption. A digital signature is created with the signer’s private key and verified with the public key; certificates bind identities to keys.

B32.1 Integrity. Confidentiality prevents disclosure.

B32.2 Alice’s public key.

B32.3 Make identical passwords produce different stored hashes and defeat precomputed lookup tables; it is normally not secret.

B32.4 Phishing. Verify through a trusted channel rather than the message’s link.

B32.5 Sign with the sender's private key and verify with the corresponding public key, using a signature algorithm. Do not confuse this with encrypting data for the receiver.

B32.6 Phishing. The central mechanism is deception to obtain credentials; a password's length does not prevent entry into a fraudulent form.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B33 • ORIGINAL PRACTICE

Linux: commands, permissions and redirection

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q168; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q135-136

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B33.1 What permissions does chmod 640 file give?

B33.2 What does grep "error" log.txt do?

B33.3 Difference between command > out.txt and command >> out.txt?

B33.4 Additional practice: Interpret Unix permission 640 for owner,group,others.

B33.5 Additional practice: What differs between > and >> in ordinary shell output redirection?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

r=4,w=2,x=1 for owner/group/others. > overwrites, >> appends, | connects output to another command. grep searches content; find searches directory trees by criteria.

B33.1 Owner read+write; group read; others none: rw-r-----.

B33.2 Prints matching lines containing the case-sensitive pattern error by default.

B33.3 > replaces file contents; >> appends. Shell redirection is not a search operation.

B33.4 Owner read+write, group read, others none. Decode each octal digit with read=4, write=2, execute=1.

B33.5 > creates/truncates the target file; >> appends. Both redirect standard output unless another descriptor is specified.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B34 • ORIGINAL PRACTICE

Software lifecycle, requirements and maintenance

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q149-154; P34 · TGT Computer Science 2021 file 2 (2021) Q145-148; P37 · TGT Computer Science 2021 file 5 (2021) Q150-153

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B34.1 A team repeatedly evaluates risks before each development cycle. Which model fits?

B34.2 “Are we building the right product?” refers to verification or validation?

B34.3 Updating software to run on a new OS is what maintenance category?

B34.4 Additional practice: What is the purpose of a requirements specification?

B34.5 Additional practice: Fixing a defect discovered after release is which maintenance category?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

SRS describes required behaviour and constraints. Waterfall suits stable sequential requirements; spiral emphasises risk analysis; prototyping clarifies uncertain requirements. Verification checks conformance; validation checks user needs.

B34.1 Spiral.

B34.2 Validation.

B34.3 Adaptive. Corrective fixes defects; perfective improves capability/performance; preventive reduces future problems.

B34.4 State what the system must do and the constraints it must satisfy, in a verifiable form. It should support agreement and validation before detailed construction.

B34.5 Corrective maintenance. Adaptive responds to environmental changes; perfective improves capability/performance; preventive reduces future faults.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B35 • ORIGINAL PRACTICE

Testing, cohesion, coupling and metrics

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q152-153; P34 · TGT Computer Science 2021 file 2 (2021) Q143,149

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B35.1 Graph has 12 edges and 10 nodes, one connected component. Cyclomatic complexity?

B35.2 Input valid range is 1-100 inclusive. Give boundary-value tests.

B35.3 Two modules share global variables. Coupling category?

B35.4 All parts of a module perform one well-defined task. Cohesion type?

B35.5 Additional practice: A control-flow graph has E=10 edges, N=8 nodes and one connected component. Find cyclomatic complexity.

B35.6 Additional practice: Which is generally preferred: high cohesion and low coupling, or low cohesion and high coupling?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Black-box tests external behaviour; white-box uses code structure. Aim for high cohesion and low coupling. For one connected control-flow graph: cyclomatic complexity=E-N+2, often decisions+1 for binary decisions.

B35.1 Four. 12-10+2.

B35.2 0,1,2,99,100,101 cover values immediately around both boundaries.

B35.3 Common coupling.

B35.4 Functional cohesion, normally a desirable strong form.

B35.5 V=E−N+2=4. This applies to the stated single connected flow graph; the general form is E−N+2P.

B35.6 High cohesion and low coupling. Keep a module's responsibilities related and reduce unnecessary dependencies between modules.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B36 • ORIGINAL PRACTICE

Project scheduling and estimation

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q154; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q125,173

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B36.1 PERT estimates are 2,5,8 days. Expected time?

B36.2 Paths A-B-D take 3+5+2 days; A-C-D take 3+4+2 days. Critical path and duration?

B36.3 Basic COCOMO organic effort uses E=2.4(KLOC)^1.05 person-months. At KLOC=1, effort?

B36.4 Additional practice: A PERT activity has optimistic 2, most likely 5 and pessimistic 8 days. Find expected duration.

B36.5 Additional practice: A critical path takes 12 days; a separate path takes 9. What is the project duration before resource constraints?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

PERT expected time=(optimistic+4×most likely+pessimistic)/6. Critical path is the longest-duration dependency path; its delay delays the project absent corrective action. Do not choose the shortest path.

B36.1 Five days. (2+20+8)/6.

B36.2 A-B-D,10 days.

B36.3 2.4 person-months. Use coefficients supplied in the question; do not confuse effort with elapsed time.

B36.4 (2+4×5+8)/6=5 days. The most-likely estimate receives weight 4 in the standard PERT formula.

B36.5 12 days. The longest dependency path controls completion, not the shortest path or the sum of all paths.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B37 • ORIGINAL PRACTICE

HTML, CSS and front-end concepts

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q182-183; P37 · TGT Computer Science 2021 file 5 (2021) Q144,184; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q107,149

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B37.1 How many HTML heading levels are there?

B37.2 Which table-cell attribute spans three columns?

B37.3 An element matches p, .note and #intro rules with equal origin and importance. Which selector is most specific?

B37.4 Additional practice: Which HTML element represents a navigation section: nav or span?

B37.5 Additional practice: With content width 200 px, padding 10 px each side and borders 2 px each side, what is total border-box width under content-box sizing?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

HTML structures content, CSS styles it, JavaScript adds behaviour. CSS selector specificity matters: inline usually beats ID, which beats class, which beats element, subject to cascade rules such as !important.

B37.1 Six: h1 through h6.

B37.2 colspan="3".

B37.3 #intro (ID). This is specificity, not source order.

B37.4 nav. Semantic elements describe the role of content; span is a generic inline container.

B37.5 224 px. Add content+left/right padding+left/right borders. Margins lie outside the border box.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B38 • ORIGINAL PRACTICE

JavaScript, events and asynchronous requests

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q145-146; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q119,177

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B38.1 Which event corresponds to a pointer entering an element: click or mouseover?

B38.2 What is a callback function?

B38.3 Does an asynchronous HTTP request require reloading the whole webpage?

B38.4 Additional practice: What is the difference between JavaScript == and ===?

B38.5 Additional practice: A Promise is fulfilled. Which handler normally processes its value: then's fulfilment handler or catch's rejection handler?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Events trigger handlers; a callback is passed to another function for later or delegated execution. AJAX means asynchronous JavaScript and XML, but modern requests may exchange JSON rather than XML.

B38.1 mouseover. mouseenter is another related event with different bubbling behaviour.

B38.2 A function supplied to another function to be invoked by that function.

B38.3 No. It can update part of the interface after data arrives.

B38.4 === compares without the type coercion performed by ==. For example, 0=="0" is true while 0==="0" is false.

B38.5 The fulfilment handler passed to then. catch handles rejections; later handlers can still throw or return a rejected Promise.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B39 • ORIGINAL PRACTICE

XML, DTD, parsers and legacy web syllabus

Observed: P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q140,179; supplied syllabus item 26

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B39.1 Is <a> well-formed XML?

B39.2 Which parser is usually better for streaming a huge XML file without retaining the whole tree: DOM or SAX?

B39.3 What is Microsoft IIS in the historical web-server syllabus?

B39.4 Do XML Schema and DTD offer identical datatype capabilities?

B39.5 Additional practice: In XML, can an element opened as <Name> be closed as </name>?

B39.6 Additional practice: What is the main processing distinction between DOM and SAX XML parsers?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

XML must be well formed (one root, proper nesting, matching tags, quoted attributes). Validity adds schema/DTD conformance. DOM builds a tree; SAX reports events. PWS, DHTML filters and TDC are legacy syllabus terms, not current recommendations.

B39.1 No. Tags cross rather than nest; correct nesting is <a>.

B39.2 SAX. DOM is more convenient for random tree access but holds the document model.

B39.3 Internet Information Services, a web-server platform. PWS means Personal Web Server, an older Microsoft product.

B39.4 No. XML Schema supports richer typed element/attribute definitions.

B39.5 No. XML names are case-sensitive; properly nested matching tags are required for well-formedness.

B39.6 DOM builds a tree representation; SAX reports parsing events in sequence. SAX can reduce memory use when the whole document need not be retained.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B40 • ORIGINAL PRACTICE

.NET, CLR, CTS, ASP.NET and data binding

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q167; P37 · TGT Computer Science 2021 file 5 (2021) Q171; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q154-155,178

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B40.1 Expand CLR and state its role.

B40.2 Does a JIT compiler typically translate IL to machine code before relevant execution?

B40.3 What is data binding in a user-interface framework?

B40.4 Expand WPF.

B40.5 Additional practice: What converts managed .NET intermediate code into native machine code during execution in a JIT-based runtime?

B40.6 Additional practice: What is garbage collection's central task in managed .NET execution?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

CLR executes managed .NET code; JIT compiles intermediate language for execution; CTS describes types; CLS specifies interoperability rules. ASP.NET is web-related; WPF is a Windows UI framework.

B40.1 Common Language Runtime; manages execution and services such as garbage collection for managed code.

B40.2 Yes. Do not confuse IL with C# source.

B40.3 Connecting data sources/properties to UI controls so displayed or entered data can be synchronised.

B40.4 Windows Presentation Foundation. It is not the expansion of ASP.NET.

B40.5 The JIT compiler. CLR provides the execution environment; CTS describes the common type system. Keep these acronyms' responsibilities distinct.

B40.6 Reclaim memory occupied by objects no longer reachable under the collector's rules. It does not mean resources such as file handles can always be left without deterministic disposal.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B41 • ORIGINAL PRACTICE

Graphics: transformations, colours and clipping

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q181-182; P37 · TGT Computer Science 2021 file 5 (2021) Q183; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q158

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B41.1 Scale (3,4) by factors 2 along x and 3 along y.

B41.2 Rotate (2,1) 90° counterclockwise about the origin.

B41.3 Convert normalised RGB (0.2,0.4,0.9) to CMY.

B41.4 What does clipping do?

B41.5 Additional practice: Translate point (2,3) by vector (5,−1).

B41.6 Additional practice: A point (2,1) is rotated 90° anticlockwise about the origin. Find the new point.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

2D scaling: (x,y)→(sx×x,sy×y). Translation adds offsets. Counterclockwise rotation: x′=x cosθ-y sinθ; y′=x sinθ+y cosθ. Normalised CMY=(1-R,1-G,1-B).

B41.1 (6,12).

B41.2 (-1,2). For 90°, use (x,y)→(-y,x).

B41.3 (0.8,0.6,0.1).

B41.4 Keeps the visible part within a viewing region and discards portions outside; it is not image scaling.

B41.5 (7,2). Add translation components. In homogeneous coordinates, translation is represented using an additional coordinate.

B41.6 (−1,2). Use (x,y)→(−y,x) for this stated angle and direction.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B42 • ORIGINAL PRACTICE

Graphics: rasterisation, curves, surfaces and solids

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q181; P34 · TGT Computer Science 2021 file 2 (2021) Q180,184; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q118,159,186

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B42.1 What does a Z-buffer compare to choose the visible surface at a pixel?

B42.2 A cubic Bézier curve uses how many control points?

B42.3 Distinguish boundary representation from constructive solid geometry.

B42.4 A 100×100 image uses 24 bits/pixel, no compression or header. Storage?

B42.5 Additional practice: A cubic Bézier curve uses how many control points?

B42.6 Additional practice: What does a Z-buffer store for visible-surface determination?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Raster displays use pixels. Z-buffer stores depth for visibility. Bézier curves use control points; they normally pass through the first and last, not every intermediate control point. Solid models represent volume as well as boundaries.

B42.1 Depth values of candidate surfaces.

B42.2 Four. Degree n uses n+1 control points.

B42.3 B-rep stores surfaces/edges/vertices; CSG combines primitive solids using Boolean operations.

B42.4 30,000 bytes. 100×100×24/8.

B42.5 Four. Degree n uses n+1 control points. The curve normally interpolates the first and last control points, not all intermediate controls.

B42.6 A depth value for each relevant pixel/sample. Compare candidate fragment depths according to the chosen coordinate convention; it is not a colour buffer alone.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B43 • ORIGINAL PRACTICE

Mobile computing and wireless systems

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q178-179; P37 · TGT Computer Science 2021 file 5 (2021) Q180-181; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q117,138,162

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B43.1 Expand GPRS and MANET.

B43.2 Which sensor directly measures angular velocity: gyroscope or accelerometer?

B43.3 What is a mobile agent in distributed computing?

B43.4 Additional practice: What does cellular handoff accomplish?

B43.5 Additional practice: In classic Mobile IP, what is the role of a home agent?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Handoff transfers an active connection between cells/access points. GSM,GPRS,EDGE,LTE indicate evolving cellular capabilities; avoid absolute speed claims without a standard/configuration. MANET is a mobile ad hoc network.

B43.1 General Packet Radio Service; Mobile Ad hoc Network.

B43.2 Gyroscope. An accelerometer measures acceleration.

B43.3 A software agent able to move between hosts/environments with its code/state as supported by the system.

B43.4 Transfers an active connection between serving cells/access resources as conditions change, supporting continuity of service. It differs from initial registration.

B43.5 Maintain the mobile node's location binding and forward/tunnel packets toward its care-of address. This is the classic protocol model, not a claim about every modern mobile network.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B44 • ORIGINAL PRACTICE

E-commerce, payments and business models

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q172-173; P34 · TGT Computer Science 2021 file 2 (2021) Q120,169; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q109,116,156

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B44.1 A manufacturer sells components to another manufacturer through a portal. Model?

B44.2 An individual sells a used book to another individual through a marketplace. Model?

B44.3 A site earns monthly access fees. Revenue model?

B44.4 Additional practice: An online manufacturer sells directly to individual consumers. Is this B2B or B2C?

B44.5 Additional practice: What is the purpose of a payment gateway in online commerce?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

B2B=business-to-business; B2C=business-to-consumer; C2C=consumer-to-consumer. EDI is structured electronic business-document exchange. A payment gateway facilitates online payment processing.

> **B44.1** B2B.

> **B44.2** C2C, though the platform itself may have a separate B2C service relationship.

> **B44.3** Subscription. Distinguish this from per-transaction commission.

> **B44.4** B2C. Classify by seller and buyer roles, not merely by whether payment is electronic.

> **B44.5** Facilitate secure communication and processing of payment authorisation between the merchant and payment ecosystem. It is not itself a guarantee against every fraud or refund dispute.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B45 • ORIGINAL PRACTICE

MIS, DSS, expert systems and knowledge management

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q178-179,184; P34 · TGT Computer Science 2021 file 2 (2021) Q176-177; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q137,157,187

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B45.1 A supermarket records each sale at checkout. System type?

B45.2 A manager compares “what-if” production plans using a model. System type?

B45.3 A technician’s experience-based troubleshooting skill is tacit or explicit knowledge?

B45.4 What does an inference engine do in an expert system?

B45.5 Additional practice: Which system is most suited to a what-if analysis of alternative pricing plans: TPS or DSS?

B45.6 Additional practice: An expert system combines a knowledge base with what reasoning component?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

TPS records routine transactions; MIS supplies structured management reports; DSS supports semi-structured decisions; expert systems combine a knowledge base and inference mechanism. Tacit knowledge is personal know-how; explicit knowledge is codified.

B45.1 Transaction Processing System.

B45.2 Decision Support System.

B45.3 Tacit. A written troubleshooting manual is explicit.

B45.4 Applies rules/logic to available facts to derive conclusions.

B45.5 DSS. A transaction-processing system records routine transactions; a decision-support system helps examine alternatives.

B45.6 An inference engine. The knowledge base stores domain rules/facts; the engine applies them to draw conclusions.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B46 • ORIGINAL PRACTICE

Business communication and management

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q104; P34 · TGT Computer Science 2021 file 2 (2021) Q104,186; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q102,175

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B46.1 Distinguish agenda from minutes.

B46.2 Which function compares actual performance against standards?

B46.3 A partner merely permits use of their name in the firm. Standard textbook term?

B46.4 Additional practice: Distinguish a meeting agenda from minutes.

B46.5 Additional practice: State the management functions commonly summarised as planning,organising,staffing,directing and controlling. Which compares results with standards?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Communication includes sender,message,channel,receiver and feedback; noise disrupts it. Management functions include planning,organising,staffing,directing and controlling. Minutes record what occurred; an agenda lists what is planned.

B46.1 Agenda lists proposed meeting business; minutes record proceedings and decisions.

B46.2 Controlling.

B46.3 Nominal partner. A sleeping partner generally contributes but does not actively manage.

B46.4 The agenda lists planned business before/during the meeting; minutes record proceedings, decisions and actions afterwards. Identify whether the question asks for planning or the official record.

B46.5 Controlling. It identifies deviations and supports corrective action; planning sets objectives and intended methods.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B47 • ORIGINAL PRACTICE

Financial accounting: equation and entries

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q147-148; P34 · TGT Computer Science 2021 file 2 (2021) Q141; P37 · TGT Computer Science 2021 file 5 (2021) Q146

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B47.1 An owner puts Rs 50,000 cash into a business. Which accounts change?

B47.2 Equipment costing Rs 12,000 is bought for cash. What happens to total assets immediately?

B47.3 A business earns Rs 5,000 revenue on credit. Is cash necessarily increased immediately?

B47.4 Additional practice: A business buys equipment for ₹20,000 cash. What is the effect on total assets at purchase?

B47.5 Additional practice: Under the traditional personal-account rule, whom do you debit and whom do you credit?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Assets=Liabilities+Equity. Debit increases assets/expenses; credit increases liabilities/equity/income. Every transaction preserves the equation. Profit is not the same as cash.

B47.1 Cash (asset) increases 50,000 and capital (equity) increases 50,000. Debit cash; credit capital.

B47.2 No net change. Equipment rises and cash falls equally.

B47.3 No. A receivable increases; cash arrives when collected.

B47.4 No net change: equipment +₹20,000, cash −₹20,000. This is an asset exchange; it is not an immediate expense of the entire asset price under normal capitalisation.

B47.5 Debit the receiver, credit the giver. This mnemonic must be applied only after identifying the account category correctly.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B48 • ORIGINAL PRACTICE

Accounting: depreciation, provisions and reconciliation

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q147; P34 · TGT Computer Science 2021 file 2 (2021) Q142; P37 · TGT Computer Science 2021 file 5 (2021) Q145,147

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B48.1 Asset cost Rs 60,000,residual Rs 10,000,life 5 years. Annual straight-line depreciation?

B48.2 Asset cost Rs 20,000,depreciation 10% WDV yearly. Value after 2 years?

B48.3 A cheque issued is entered in the cash book but not yet presented. Under a positive-balance assumption, is bank-statement balance temporarily higher or lower?

B48.4 Additional practice: An asset costs ₹50,000, has residual value ₹5,000 and a 5-year useful life. Find annual straight-line depreciation.

B48.5 Additional practice: A ₹10,000 asset depreciates by 20% annually using written-down value. What is book value after 2 years?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Straight-line annual depreciation=(cost-residual value)/life. Written-down value applies a rate to opening book value. A provision is a charge for an obligation/expected loss; a reserve is generally an appropriation of profit.

B48.1 Rs 10,000.

B48.2 Rs 16,200. 20000×0.9^2.

B48.3 Higher than the cash-book balance by that cheque amount, ignoring other differences.

B48.4 ₹9,000. (Cost−residual value)/useful life=(50,000−5,000)/5. Depreciable amount excludes residual value.

B48.5 ₹6,400. Multiply $10,000\times0.8^2$. Equal percentage reductions do not produce equal annual depreciation amounts.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B49 • ORIGINAL PRACTICE

Business economics and banking concepts

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q165-167; P34 · TGT Computer Science 2021 file 2 (2021) Q159-160; P37 · TGT Computer Science 2021 file 5 (2021) Q162-164; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q153,183

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B49.1 Price rises 10% and quantity demanded falls 20%. Approximate elasticity magnitude using those stated percentage changes?

B49.2 Total cost rises from Rs 500 to Rs 560 when output rises from 10 to 12. Marginal cost per extra unit?

B49.3 “The government should reduce inequality” is positive or normative?

B49.4 What distinguishes CRR from SLR conceptually?

B49.5 Additional practice: Price rises by 10% and quantity demanded falls by 20%. Find the magnitude of demand elasticity using the stated percentage-change method.

B49.6 Additional practice: What is the difference between fixed cost and marginal cost?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Demand elasticity=(% change in quantity)/(% change in price); usually quote its absolute magnitude. Marginal cost is change in total cost/change in output. Positive economics describes testable claims; normative economics makes value judgments.

B49.1 2, so demand is elastic.

B49.2 Rs 30. Change cost 60/change output 2.

B49.3 Normative. “Should” expresses a value judgment.

B49.4 CRR is the prescribed cash reserve with RBI; SLR is the prescribed liquid-asset holding. Current rates must be checked separately.

B49.5 2. $|\%\Delta Q/\%\Delta P|=20/10$. It is elastic demand. The midpoint method is a different convention unless specified.

B49.6 Fixed cost does not change with output within the relevant range; marginal cost is the extra total cost of an additional unit. $MC=\Delta TC/\Delta Q$ for a discrete change.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B50 • ORIGINAL PRACTICE

Basic physics and electronics

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q127; P34 · TGT Computer Science 2021 file 2 (2021) Q125; P37 · TGT Computer Science 2021 file 5 (2021) Q125; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q126-127,144-145

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B50.1 A 12 V source drives a 6 Ω resistor. Find current and power.

B50.2 Find equivalent resistance of 6 Ω and 3 Ω in parallel.

B50.3 Classify speed,velocity,mass and force as scalars/vectors.

B50.4 For Coulomb’s law, if distance doubles with charges unchanged, force becomes what fraction?

B50.5 Additional practice: A 12 V supply is connected across a 6 Ω resistor. Find current and power.

B50.6 Additional practice: What happens to resistance if a wire's length doubles and cross-sectional area stays fixed, at the same temperature?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

V=IR; P=VI=I²R=V²/R. Series resistances add; parallel reciprocals add. Scalar has magnitude only; vector has magnitude and direction. Physics is explicitly in the supplied subject syllabus.

B50.1 2 A and 24 W. I=12/6; P=12×2.

B50.2 2 Ω. Product/sum=18/9.

B50.3 Speed and mass are scalars; velocity and force are vectors.

B50.4 One-quarter. Force varies as $1/r^2$.

B50.5 2 A and 24 W. I=V/R and P=VI. Check that resistance is in ohms, not kilohms.

B50.6 It doubles. R=ρL/A. A wire stretched to conserve volume is a different problem because its area also changes.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B51 • ORIGINAL PRACTICE

Mathematics: sets, logic and matrices

Observed: P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q121,141-142; scope within broad Mathematics I-IV

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B51.1 A={1,2,3},B={3,4}. Find union and intersection.

B51.2 Find determinant of [[2,1],[3,4]].

B51.3 Negate “All programs terminate.”

B51.4 If A is 2×3 and B is 3×4, what is the size of AB?

B51.5 Additional practice: Find the determinant of [[2,3],[1,4]].

B51.6 Additional practice: Negate “All students passed.”

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

|A∪B|=|A|+|B|-|A∩B|. A relation need not be a function; a function gives exactly one output per input. For [[a,b],[c,d]], determinant=ad-bc. Matrix multiplication is generally not commutative.

B51.1 Union={1,2,3,4}; intersection={3}.

B51.2 5. 2×4-1×3.

B51.3 “At least one program does not terminate.” Negating “all” gives “some not”, not “none”.

B51.4 2×4. Inner dimensions agree; outer dimensions remain.

B51.5 5. For a 2×2 matrix use ad−bc=8−3. A nonzero determinant means this square matrix is invertible.

B51.6 “At least one student did not pass.” Negating ∀x P(x) gives ∃x ¬P(x), not “No student passed”.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B52 • ORIGINAL PRACTICE

Mathematics: calculus and differential equations

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q101; P34 · TGT Computer Science 2021 file 2 (2021) Q101; broad mathematics extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B52.1 Find d/dx (3x²+2x-1).

B52.2 Find the minimum of x² on [-2,3].

B52.3 State order and degree of (d²y/dx²)³+dy/dx+y=0.

B52.4 Evaluate $\int_0^2 x\,dx$.

B52.5 Additional practice: Differentiate x²e^x.

B52.6 Additional practice: Solve dy/dx=3y with y(0)=2.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Differentiate x^n as n x^(n-1). For an interval minimum, test stationary points AND endpoints. Order of a differential equation is the highest derivative order; degree requires polynomial form in derivatives.

B52.1 6x+2.

B52.2 0 at x=0. Endpoints give 4 and 9.

B52.3 Order 2,degree 3.

B52.4 2. Antiderivative x²/2; at 2 minus at 0.

B52.5 e^x(x²+2x). Apply the product rule; differentiating only one factor omits a term.

B52.6 y=2e^(3x). Separate dy/y=3dx, integrate, and use the initial condition to fix the constant.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B53 • ORIGINAL PRACTICE

Statistics: averages, dispersion and regression

Observed: P34 · TGT Computer Science 2021 file 2 (2021) Q189; P37 · TGT Computer Science 2021 file 5 (2021) Q102,187-188

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B53.1 Find mean,median and mode of 2,3,3,8.

B53.2 For population data 1,3,find variance and standard deviation.

B53.3 If X has mean 5 and variance 4, find mean and variance of Y=3X+2.

B53.4 If correlation is r=0.6,SD(Y)=10,SD(X)=5, regression slope of Y on X?

B53.5 Additional practice: Data are 2,4,6. Find the mean and population variance.

B53.6 Additional practice: Correlation r=0.8. What is r^2 and its usual simple-linear-regression interpretation with an intercept?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Sort data before finding median. Population variance divides by n; unbiased sample variance by n-1. For Y=a+bX: mean transforms likewise, variance multiplies by b^2. Correlation does not prove causation.

B53.1 Mean=4; median=3; mode=3.

B53.2 Mean=2; squared deviations=1,1; variance=1; SD=1.

B53.3 Mean=17; variance=36. Adding 2 does not affect variance.

B53.4 $b_{yx}=r\times SD(Y)/SD(X)=1.2$.

B53.5 Mean=4, variance=$[(-2)^2+0^2+2^2]/3=8/3$. A sample variance would divide by n−1 and give 4.

B53.6 0.64: 64% of the sample variation in the dependent variable is explained by the fitted linear model in that setting. It does not prove causation.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B54 • ORIGINAL PRACTICE

Probability, conditional probability and Bayes

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q189; P34 · TGT Computer Science 2021 file 2 (2021) Q190; P37 · TGT Computer Science 2021 file 5 (2021) Q103,189

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B54.1 Two independent people solve a problem with probabilities 1/2 and 1/3. Probability at least one solves?

B54.2 A bag has 3 red and 2 blue balls. Two drawn without replacement. Probability both red?

B54.3 Machines A,B supply 60%,40% of items; defect rates 2%,5%. A defective item is found. Probability it came from B?

B54.4 Additional practice: Draw one card from a standard 52-card deck. Find P(king or heart).

B54.5 Additional practice: Disease prevalence is 1%; test sensitivity 90%; false-positive rate 5%. Find P(disease|positive).

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

For independent events multiply probabilities. For “at least one”, use 1-P(none). Conditional probability changes the sample space. Bayes posterior ∝ prior×likelihood.

B54.1 1-(1/2)(2/3)=2/3.

B54.2 (3/5)(2/4)=3/10. With replacement it would be 9/25.

B54.3 0.4×0.05/(0.6×0.02+0.4×0.05)=0.02/0.032=5/8.

B54.4 (4+13−1)/52=4/13. Subtract the king of hearts counted in both sets.

B54.5 0.009/(0.009+0.0495)=2/13≈15.38%. Use a base of 10,000 people: 90 true positives and 495 false positives. A high sensitivity is not the same as this posterior probability.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B55 • ORIGINAL PRACTICE

Distributions and statistical inference

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q188; inference is explicitly in syllabus item 30

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B55.1 For Binomial(10,0.3), find mean and variance.

B55.2 Poisson mean is 9. Find standard deviation.

B55.3 Population SD is 12 and n=36 independent observations. Standard error of sample mean?

B55.4 At a preselected significance level 0.05, p=0.03. Decision?

B55.5 Additional practice: For X~Binomial(n=10,p=0.3), give mean and variance.

B55.6 Additional practice: For a standard normal variable, what approximate probability lies between −1 and +1?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Binomial(n,p): mean=np,variance=np(1-p). Poisson(λ): mean=variance=λ. Standard error of sample mean=σ/√n under usual independent-sampling assumptions. A p-value is not the probability the null hypothesis is true.

B55.1 Mean=3; variance=2.1.

B55.2 3. Variance=9; SD=√9.

B55.3 2. 12/√36.

B55.4 Reject the null under that testing rule; this does not measure practical importance or prove the alternative.

B55.5 Mean=np=3; variance=np(1−p)=2.1. Standard deviation is √2.1, not 2.1.

B55.6 About 0.6827. This is a distribution fact; the wider ±2 and ±3 intervals have different probabilities.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B56 • ORIGINAL PRACTICE

Interpolation and numerical methods

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q190; P36 · TGT Computer Science 2021 file 4 (2021) Q190; P37 · TGT Computer Science 2021 file 5 (2021) Q190

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B56.1 Given $f(2)=5, f(6)=13$, estimate $f(3)$ by linear interpolation.

B56.2 For $f(x)=x^2-2$, start $x_0=1$. One Newton-Raphson step?

B56.3 Use trapezoidal rule with one interval to approximate $\int_0^2 x^2\,dx$.

B56.4 Data points (0,1),(1,2),(2,5) lie on which degree-at-most-two interpolating polynomial?

B56.5 Additional practice: Linearly interpolate y at x=3 between (2,5) and (4,9).

B56.6 Additional practice: For $f(x)=x^2-2$, take one Newton-Raphson step from $x_0=1$.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Two-point linear interpolation: $y=y_0+(x-x_0)(y_1-y_0)/(x_1-x_0)$. Newton forward/backward require equally spaced nodes in their usual finite-difference forms. Lagrange permits unequal spacing. Interpolation estimates inside the data range.

B56.1 7. Slope=(13-5)/4=2; 5+2×1.

B56.2 1.5. $x_1=x_0-f(x_0)/f'(x_0)=1-(-1)/2$.

B56.3 4. Width 2 times average of endpoint values (0+4)/2. Exact value 8/3 differs.

B56.4 x^2+1. Substitute each point; uniqueness holds for three distinct x-values.

B56.5 7. $y=5+(3-2)(9-5)/(4-2)$. It is the midpoint in x, so it is the midpoint in y under linear interpolation.

B56.6 $x_1=1-(1-2)/2=1.5$. Use $x_{next}=x-f(x)/f'(x)$; a zero derivative would make this step invalid.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B57 • ORIGINAL PRACTICE

Advanced mathematics reserve: complex numbers and transforms

Observed advanced-maths prompts: P33 · TGT Computer Science 2021 file 1 (2021) Q103; P34 · TGT Computer Science 2021 file 2 (2021) Q102-103. Not a complete undergraduate maths course.

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B57.1 Find (2+3i)(1-i).

B57.2 Find the modulus of 3-4i.

B57.3 What is the Laplace transform of 1 for s>0?

B57.4 For an analytic numerator g, what is $\oint g(z)/(z-a)\,dz$ around a positively oriented simple closed contour containing a once?

B57.5 Additional practice: Find the modulus of 3−4i.

B57.6 Additional practice: Give the Laplace transform of 1 for s>0.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

The supplied Mathematics I-IV headings are underspecified. Treat this as a reserve drill after core topics. $i^2=-1$; modulus of a+bi is $\sqrt{(a^2+b^2)}$. Standard Laplace transforms use suitable convergence conditions.

B57.1 5+i. Expand: 2-2i+3i-3i².

B57.2 5. √(9+16).

B57.3 1/s. Integrate e^(-st) from 0 to infinity.

B57.4 2πi g(a), assuming g is analytic on and inside the contour (Cauchy integral formula).

B57.5 5. |a+bi|=√(a²+b²); do not subtract the squared imaginary part when calculating modulus.

B57.6 1/s. Integrate e^(−st) from 0 to ∞. The condition ensures convergence for the ordinary transform.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B58 • ORIGINAL PRACTICE

Teaching: learner-centred methods and learning theories

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q191; P34 · TGT Computer Science 2021 file 2 (2021) Q191; P36 · TGT Computer Science 2021 file 4 (2021) Q191; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q194-195

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B58.1 Students discover a sorting rule through examples and discuss their reasoning. Teacher-centred or learner-centred?

B58.2 A teacher first models a task, then gradually removes support. What is this?

B58.3 In revised Bloom taxonomy, “design a new algorithm” most directly illustrates which level?

B58.4 Additional practice: A teacher gives progressively smaller hints as a learner gains independence. Which concept does this illustrate?

B58.5 Additional practice: A lesson begins with pupils predicting an algorithm's output, testing it and explaining their errors. Which approach is emphasised?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Choose methods that actively reveal understanding. Constructivism emphasises learners building understanding; scaffolding is temporary support; ZPD concerns what a learner can do with guidance beyond independent ability.

B58.1 Learner-centred: students actively construct and explain the rule.

B58.2 Scaffolding, followed by gradual release of responsibility.

B58.3 Create. Remember recalls; understand explains; apply uses; analyse examines; evaluate judges.

B58.4 Scaffolding. Temporary support is adjusted and faded; supplying the complete answer every time does not build the same independence.

B58.5 Active, constructive learning. The learners use prior ideas, evidence and reflection; the important feature is their thinking, not merely the presence of a computer.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B59 • ORIGINAL PRACTICE

Teaching: assessment, feedback and test quality

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q193-194; P34 · TGT Computer Science 2021 file 2 (2021) Q193-194; P36 · TGT Computer Science 2021 file 4 (2021) Q193-194; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q196-197

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B59.1 A teacher uses an exit ticket to change tomorrow's lesson. Assessment type?

B59.2 A test repeatedly gives similar scores but measures spelling when intended to measure programming skill. Reliable and valid?

B59.3 Best feedback: "Wrong", or "Your loop stops before the last item; trace the boundary and revise it"?

B59.4 An MCQ with a single correct alternative is open-ended or closed-ended?

B59.5 Additional practice: A short quiz is used during teaching to decide what to reteach. Is its use formative or summative?

B59.6 Additional practice: A test repeatedly gives consistent scores but does not measure the intended learning outcome. Is it necessarily valid?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Assessment for learning is formative; of learning is summative; as learning emphasises learner self-monitoring. Diagnostic assessment locates prior knowledge/difficulties. Reliability=consistency; validity=appropriate interpretation for the intended use.

B59.1 Formative, because evidence guides ongoing instruction.

B59.2 It may be reliable but lacks validity for that intended programming interpretation.

B59.3 The specific second response. It identifies the issue and a next step.

B59.4 Closed-ended.

B59.5 Formative. Classification depends on the purpose and use of feedback, not just on whether the quiz has marks.

B59.6 No. Consistency supports reliability, while validity concerns the interpretation/use for the intended construct. Reliability alone cannot establish validity.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B60 • ORIGINAL PRACTICE

Teaching: inclusion, language and gender

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q195-200; P34 · TGT Computer Science 2021 file 2 (2021) Q195-200; P37 · TGT Computer Science 2021 file 5 (2021) Q195-200

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B60.1 A learner struggles to read small text on a programming worksheet. Appropriate first adaptation?

B60.2 Does assigning all coding to boys and all documentation to girls promote equality?

B60.3 Which better supports language learning: meaningful dialogue or only copying isolated word lists?

B60.4 Additional practice: A pupil with low vision struggles to read projected code. What is an appropriate first instructional adjustment?

B60.5 Additional practice: A teacher repeatedly assigns coding to boys and documentation to girls. What should change?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Support access without stereotyping. Language acquisition grows through meaningful interaction; multilingual experience can support learning. Accommodations remove barriers without automatically changing the assessed construct.

B60.1 Provide accessible enlarged/high-contrast text or compatible digital material, based on their needs.

B60.2 No. Rotate roles and provide equal opportunities based on learning goals rather than gender.

B60.3 Meaningful dialogue supports communicative use; vocabulary practice can supplement it.

B60.4 Provide accessible large/high-contrast text and a usable digital copy, following the pupil's needs. Preserve the learning objective while changing access to the material.

B60.5 Rotate roles and offer equitable opportunities based on learning needs and interests. The original arrangement reinforces gender stereotypes instead of developing everyone's skills.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B61 • ORIGINAL PRACTICE

Teaching: aids, arts, ICT and classroom application

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q192,198; P36 · TGT Computer Science 2021 file 4 (2021) Q192,198; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q198

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B61.1 A dangerous experiment cannot be performed safely in class. Suitable aid?

B61.2 Students act out packet routing and discuss congestion. Which benefits can this provide?

B61.3 How can local arts support teaching?

B61.4 Additional practice: A school has intermittent internet. How can a coding lesson still include all pupils?

B61.5 Additional practice: When choosing a teaching aid, what should be checked first: visual novelty or fit with the learning objective?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Choose teaching aids by objective, safety, accessibility and cost. Audio aids include radio; visual aids include maps; video is audiovisual. Use assessment evidence to adapt instruction, not merely to rank students.

B61.1 A well-chosen simulation or demonstration video, with discussion of its limitations.

B61.2 Active modelling, collaboration and conceptual discussion; follow it with questions checking individual understanding.

B61.3 Connect concepts to learners' cultural contexts and provide multiple forms of expression without stereotyping communities.

B61.4 Prepare offline examples/tools and an unplugged tracing option aligned with the same objective. Choose the teaching method around the learning goal and available access.

B61.5 Fit with the objective and learner needs. Then assess accuracy, accessibility and practicality; novelty alone is not an instructional justification.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B62 • ORIGINAL PRACTICE

PC software, cloud computing and open source

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q101-106; P36 · TGT Computer Science 2021 file 4 (2021) Q103

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B62.1 Copy =$A$1+B2 from C2 to C3. What formula results?

B62.2 Which cloud model provides ready-to-use applications: IaaS,PaaS or SaaS?

B62.3 Does open-source software necessarily cost zero and permit every use without conditions?

B62.4 Which Word shortcuts justify a paragraph and undo an action?

B62.5 Additional practice: In a spreadsheet, what happens to =$A1+B$2 when copied one cell right and one cell down?

B62.6 Additional practice: Which cloud model supplies virtual machines for customers to configure: IaaS or SaaS?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Common Windows Word shortcuts: Ctrl+Z undo,Ctrl+H replace,Ctrl+J justify,Ctrl+T hanging indent. In spreadsheet formulas $ fixes a row/column. Cloud service models differ in what the customer manages.

B62.1 =$A$1+B3. Absolute A1 stays fixed; relative B2 moves one row.

B62.2 SaaS. IaaS offers infrastructure; PaaS offers a managed application platform.

B62.3 No. Licence conditions govern use, modification and redistribution; price and source availability are separate concepts.

B62.4 Ctrl+J and Ctrl+Z, respectively, in the common Windows mapping.

B62.5 =$A2+C$2. $A fixes the first reference's column; $2 fixes the second reference's row. Shift only the unfixed components.

B62.6 IaaS. SaaS supplies an application service; IaaS gives infrastructure resources with more configuration responsibility.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B63 • ORIGINAL PRACTICE

Algorithm traces: sorting passes and round robin

Observed: P36 · TGT Computer Science 2021 file 4 (2021) Q110,151,172-173

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B63.1 Perform one left-to-right ascending bubble pass on [5,1,4,2].

B63.2 Perform two ascending selection-sort passes on [5,1,4,2].

B63.3 Stable LSD radix sort of [31,14,22,13] after the units-digit pass?

B63.4 Round robin quantum 2; A burst 5,B burst 3; both arrive at zero, A first. Completion times?

B63.5 Additional practice: After the first left-to-right bubble-sort pass on [5,1,4,2], using adjacent ascending swaps, what is the array?

B63.6 Additional practice: Two jobs arrive at time 0 in order A,B with bursts 3 and 2. Round-robin quantum=2, no overhead. Give completion times.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

State the precise algorithm variant. A bubble pass left-to-right moves the largest remaining item right; selection sort picks the minimum; stable LSD radix sort processes units before tens.

B63.1 [1,4,2,5]. Compare adjacent pairs and swap 5 repeatedly.

B63.2 [1,2,4,5]. First place 1, then place 2.

B63.3 [31,22,13,14], ordered by final digits 1,2,3,4.

B63.4 A=8,B=7. Timeline A0-2,B2-4,A4-6,B6-7,A7-8. Waiting times: A=3,B=4.

B63.5 [1,4,2,5]. The largest element moves to the end in this pass. Do not report the fully sorted array after only one pass.

B63.6 A completes at 5, B at 4. Schedule A 0–2, B 2–4, A 4–5. Completion time differs from waiting time.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B64 • ORIGINAL PRACTICE

Flowcharts, reliability and distributed databases

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q186; P35 · TGT Computer Science 2021 file 3 (2021) Q132-133,141; P36 · TGT Computer Science 2021 file 4 (2021) Q130,186-187

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B64.1 Which flowchart symbol tests whether x>0?

B64.2 MTTF=90 hours,MTTR=10 hours. Availability?

B64.3 A table is divided by rows into North and South region records. Horizontal or vertical fragmentation?

B64.4 What is a checkpoint used for in recovery?

B64.5 Additional practice: A series system has independent component reliabilities 0.9 and 0.8. Find system reliability.

B64.6 Additional practice: Distinguish fragmentation from replication in distributed databases.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Flowchart oval=start/end,rectangle=process,diamond=decision,parallelogram=input/output. Availability=MTTF/(MTTF+MTTR) under a simple repair-cycle model. Fragmentation divides a database; allocation places fragments.

B64.1 Diamond, with branches for outcomes.

B64.2 90%. 90/(90+10).

B64.3 Horizontal. Vertical fragmentation divides attributes, usually retaining key attributes for reconstruction.

B64.4 Records a recovery reference point so recovery need not process the entire history from the beginning.

B64.5 0.72. Both must work, so multiply under the independence assumption. A parallel redundant system has a different formula.

B64.6 Fragmentation divides data into parts; replication keeps copies at multiple sites. Both may coexist; storing copies is not itself horizontal fragmentation.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B65 • ORIGINAL PRACTICE

Physics reserve: fluids, elasticity and units

Observed: P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q126-127,144-145

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B65.1 A wire has stress 2×10^8 Pa and strain 0.001. Young’s modulus?

B65.2 A liquid’s path becomes irregular with eddies. Laminar or turbulent?

B65.3 Which pair contains vectors: force and displacement, or mass and temperature?

B65.4 Additional practice: A pressure of 200 Pa acts uniformly on 0.5 m². Find force.

B65.5 Additional practice: A wire has stress 2×10^8 Pa and strain 10^−3 in its elastic linear range. Find Young's modulus.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Laminar flow is orderly; turbulent flow is irregular. Young’s modulus=longitudinal stress/longitudinal strain; bulk modulus relates pressure change to volumetric strain. Stress units are pascals; strain is dimensionless.

B65.1 2×10^11 Pa. Divide stress by strain.

B65.2 Turbulent.

B65.3 Force and displacement. Direction matters for both.

B65.4 100 N. F=PA. Use area in square metres when pressure is in pascals.

B65.5 2×10^11 Pa. Modulus=stress/strain; strain is dimensionless. The proportional law is applied within its stated range.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

B66 • ORIGINAL PRACTICE

Numerical reserve: finite differences and integration

Observed: P36 · TGT Computer Science 2021 file 4 (2021) Q190; syllabus numerical techniques/interpolation

QUESTIONS FIRST · PATTERN-BASED PRACTICE

B66.1 Values at x=0,1,2 are y=1,3,7. First and second forward differences?

B66.2 Using these data, estimate y(0.5) with the quadratic Newton forward polynomial.

B66.3 Use Simpson's 1/3 rule on x^2 from 0 to 2 with h=1.

B66.4 Additional practice: Use the trapezoidal rule with one interval to approximate $\int_0^2 x^2\,dx$.

B66.5 Additional practice: For equally spaced values $y_0=1, y_1=4, y_2=9$, find Δy_0 and $\Delta^2 y_0$.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

For equally spaced nodes, Newton forward uses $p=(x-x_0)/h$; backward uses $p=(x-x_n)/h$. Linear interpolation is exact only for a line. Simpson's 1/3 rule needs an even number of equal subintervals.

B66.1 First: 2,4. Second: 2.

B66.2 1.75. p=0.5: 1+p×2+p(p-1)×2/2=1+1-0.25.

B66.3 (1/3)[0+4×1+4]=8/3, exact for this quadratic.

B66.4 4. h/2[f(0)+f(2)]=2/2(0+4). The exact integral is 8/3; a numerical approximation need not equal it.

B66.5 3 and 2. First differences are 3,5; the second difference is 5−3=2. Preserve the indexing order.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

Verified PYQ bridge: worked technical items

Official paper references P33 · TGT Computer Science 2021 file 1 (2021) Q190; P34 · TGT Computer Science 2021 file 2 (2021) Q115,154; P37 · TGT Computer Science 2021 file 5 (2021) Q120

QUESTIONS FIRST · PATTERN-BASED PRACTICE

BP1.1 P33 Q190: With f(1)=4 and f(5)=6, estimate f(3) by two-point interpolation.

BP1.2 P34 Q115: What are the modern C remainders 7%-3 and -7%3?

BP1.3 P34 Q154: Three empty frames; references 4,7,6,1,7,6,1,2,7,2. Optimal faults?

BP1.4 P37 Q120: Convert $(84)_{12}$ to base 8.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

These are shortened paraphrases of verified past-paper items. Explanations are independently worked; historical terminology is retained only where needed to understand the item.

BP1.1 5. x=3 is the midpoint, so the linear interpolated value is the average (4+6)/2.

BP1.2 1 and -1. Truncation toward zero gives quotient -2 in each case; remainder retains dividend sign.

BP1.3 Five. Initial three loads; at 1 evict 4; at 2 evict 6 or 1, neither used again. Other references hit.

BP1.4 $(144)_8$. Decimal=8×12+4=100; 100=1×64+4×8+4.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

Sources and further checks

Paper IDs refer to the official files in the 30-day plan’s source register. A question number is the number printed in that file, not a coaching-site serial number. Old answer-key colours may not survive text copying. Open the PDF to inspect the original.

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P34 · TGT Computer Science 2021 file 2

P35 · TGT Computer Science 2021 file 3

P36 · TGT Computer Science 2021 file 4

P37 · TGT Computer Science 2021 file 5

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